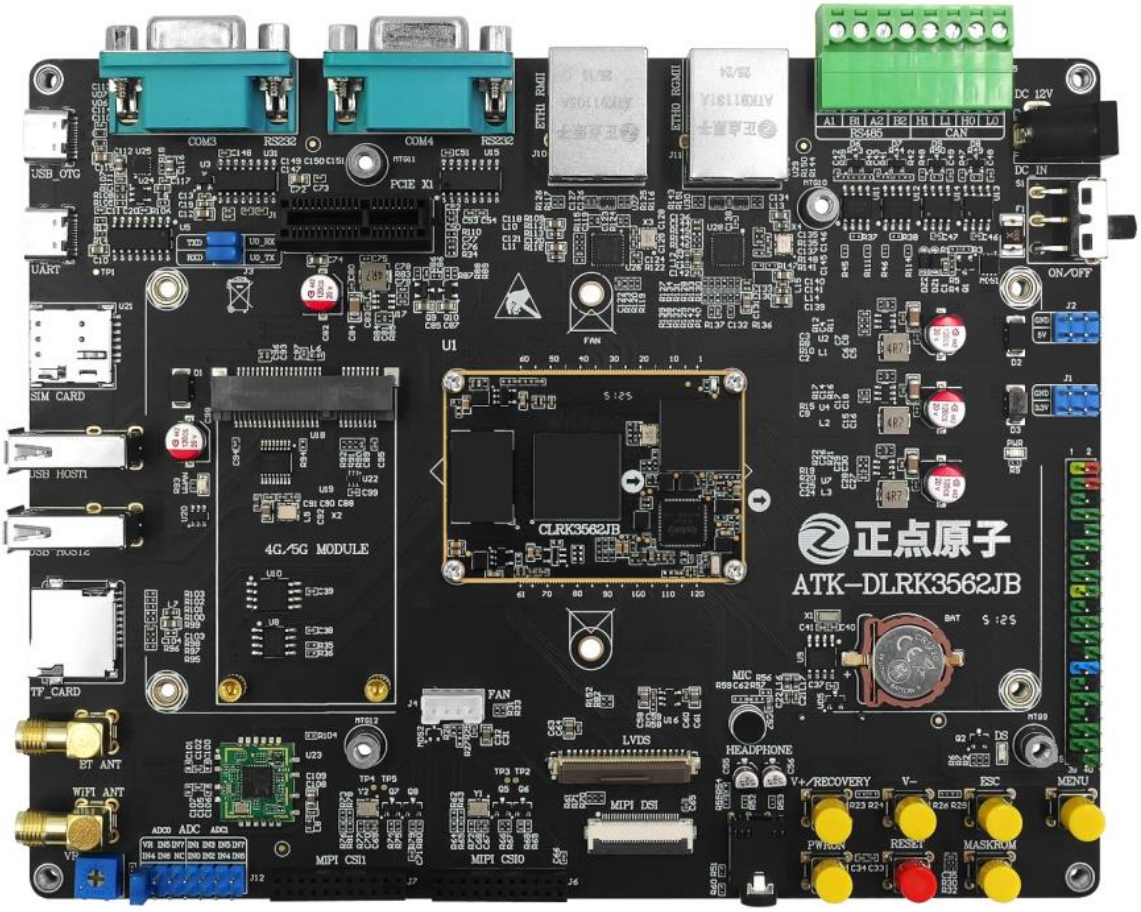
ATK-DLRK3562JB

Development Board Specification

V1.0



ALIENTEK

1. Shopping:

TMALL: https://zhengdianyuanzi.tmall.com

TAOBAO: https://openedv.taobao.com

2. Download

Address: http://www.openedv.com/docs/index.html

3. FAE

Website : www.alientek.com

Forum : http://www.openedv.com/forum.php

Videos : www.yuanzige.com

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In order to get the latest version of product information, please regularly visit the download center or contact the customer service of Taobao ALIENTEK flagship store. Thank you for your tolerance and support.

Revision History:

Version	Version Update Notes	Responsible person	Proofreading	Date
V1.0	release officially	ALIENTEK Linux Team	ALIENTEK Linux Team	2026.03

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Chapter 1. Overview of Development Board

1.1 Introduction to the ATK-DLRK3562JB Development Board

The ALIENTEK ATK-DLRK3562JB development board is a high-performance embedded development platform based on the Rockchip RK3562J industrial-grade processor, equipped with abundant hardware interfaces and comprehensive software and hardware support. The development board consists of the ATK-CLRK3562JB core board and baseboard, connected via a board-to-board (BTB) connector. The core board is standardly configured with LPDDR4/4X high-speed memory and large-capacity eMMC storage, meeting the capacity requirements of most product development. Different memory and storage capacities can also be customized according to mass production demands.

The ALIENTEK ATK-CLRK3562JB core board adopts the Rockchip RK3562J industrial-grade high-performance and low-power processor. It features a heterogeneous architecture with quad-core ARM Cortex-A53 and single-core Cortex-M0, delivering a main frequency up to 1.8 GHz. It integrates a 1.0 TOPS NPU, Mali-G52-2EE GPU and 13M ISP. The processor supports operating systems such as Linux and Android to meet the requirements of complex applications. Meanwhile, it integrates rich peripheral interfaces, including UART, I2C, I2S, SPI, CAN, SDMMC, LVDS / MIPI DSI, MIPI CSI, USB 2.0, USB 3.0 / PCIE 2.1, ADC, 100Mbps Ethernet and Gigabit Ethernet. It is widely applied in industrial HMI, data acquisition terminals, IoT edge computing, smart home, smart white appliances, building control, smart commercial display, electronic storage lockers, new energy, power control, industrial control, industrial gateways and other fields.

The baseboard of the development board is equipped with extensive peripheral resources, including 1 × Gigabit Ethernet, 1 × 100Mbps Ethernet, 2 × RS232, 2 × RS485, 2 × CAN, 2 × USB 2.0 HOST, 1 × PCIE 2.0, 1 × MIPI DSI, 2 × MIPI CSI, 1 × LVDS, 1 × Wi-Fi & Bluetooth, 1 × USB OTG, 1 × TF card slot and audio interfaces. It also supports expansion modules such as 4G module, PCIE to M.2 NVMe SSD, and PCIE to Gigabit Ethernet adapter. It is highly suitable for embedded development evaluation and product application integration.

The development board is supplied with abundant development resources.

Software resources include but are not limited to U-Boot source code, Linux kernel source code, peripheral driver source code, file system source code, Qt interface source code, and Linux application source code, as well as relevant development tools and development environments.

Document resources consist of detailed development documents and auxiliary materials, including dozens of manuals such as the SDK development manual, embedded Linux driver development guide, Linux application hardware development examples, and embedded AI development manual. The documents are easy to understand and highly practical.

Hardware resources are comprehensive, covering the core board specification, hardware reference manual, minimal system hardware design reference, baseboard schematic and original project files, core board interface datasheet, baseboard component footprint library, mechanical dimension drawings, and chip reference manuals, fully meeting developers' usage requirements.

1.2 Application Areas

Widely applicable to industrial human-machine interfaces, data acquisition terminals, IoT edge computing, smart home, smart white appliances, building control, smart commercial display, electronic storage lockers, new energy, power control, industrial control, industrial gateways and other fields.

Chapter 2. Development Board Resources

2.1 Hardware Resources

The baseboard of the ATK-DLRK3562JB development board is equipped with extremely rich hardware resources, giving full play to all internal capabilities of the RK3562J processor. Almost all built-in resources of the processor can be verified on this development board. Meanwhile, it is expanded with abundant interfaces and functional modules, making it highly suitable for user evaluation and application development.

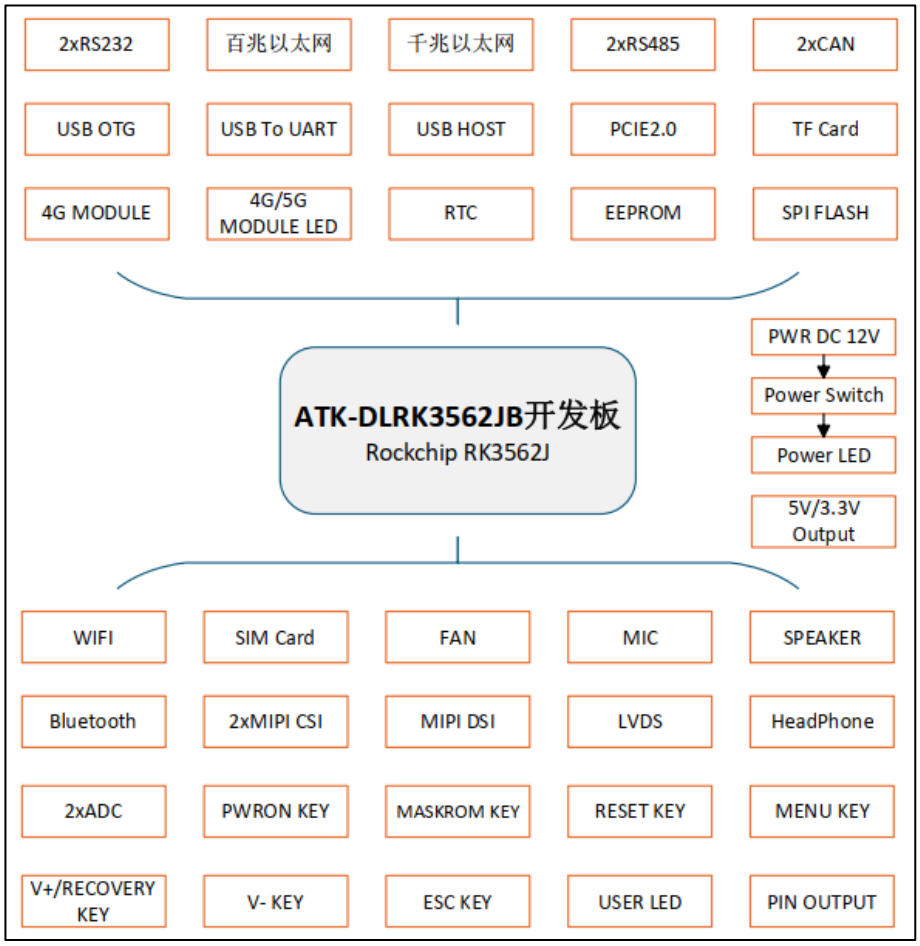


Figure 2.1-1 Development Board Hardware Resource Block Diagram

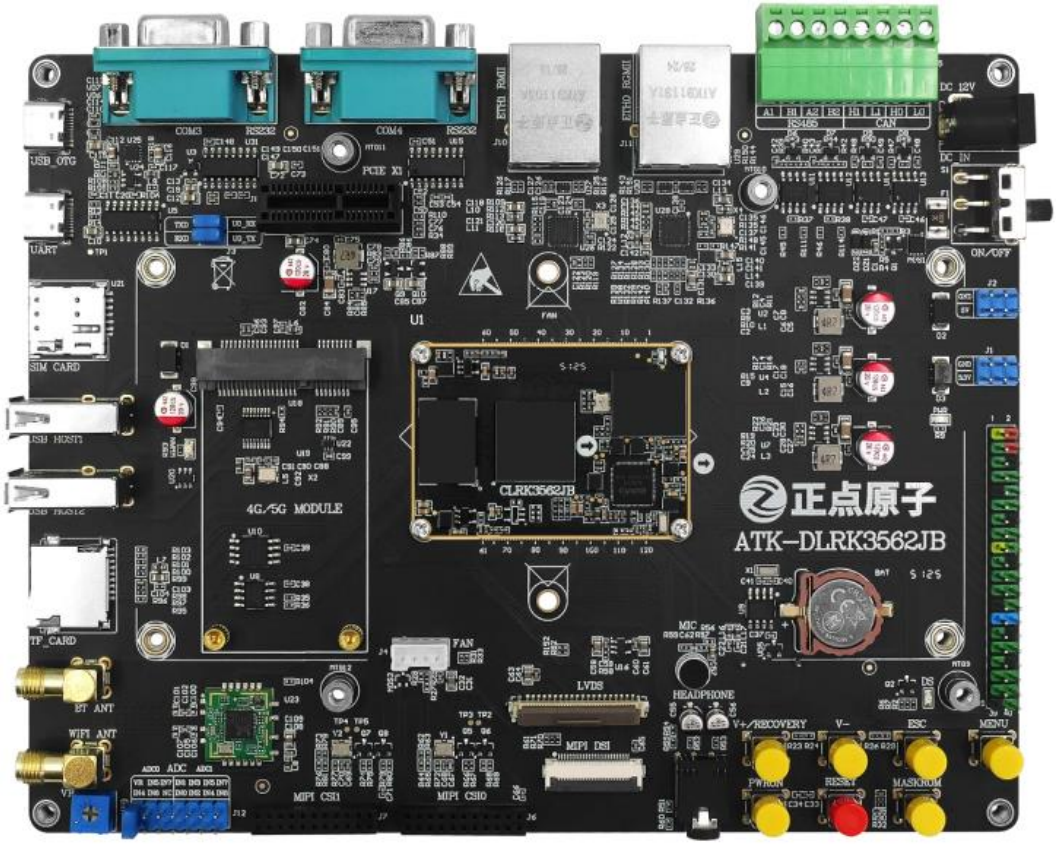


Figure 2.1-2 Front Side of Development Board

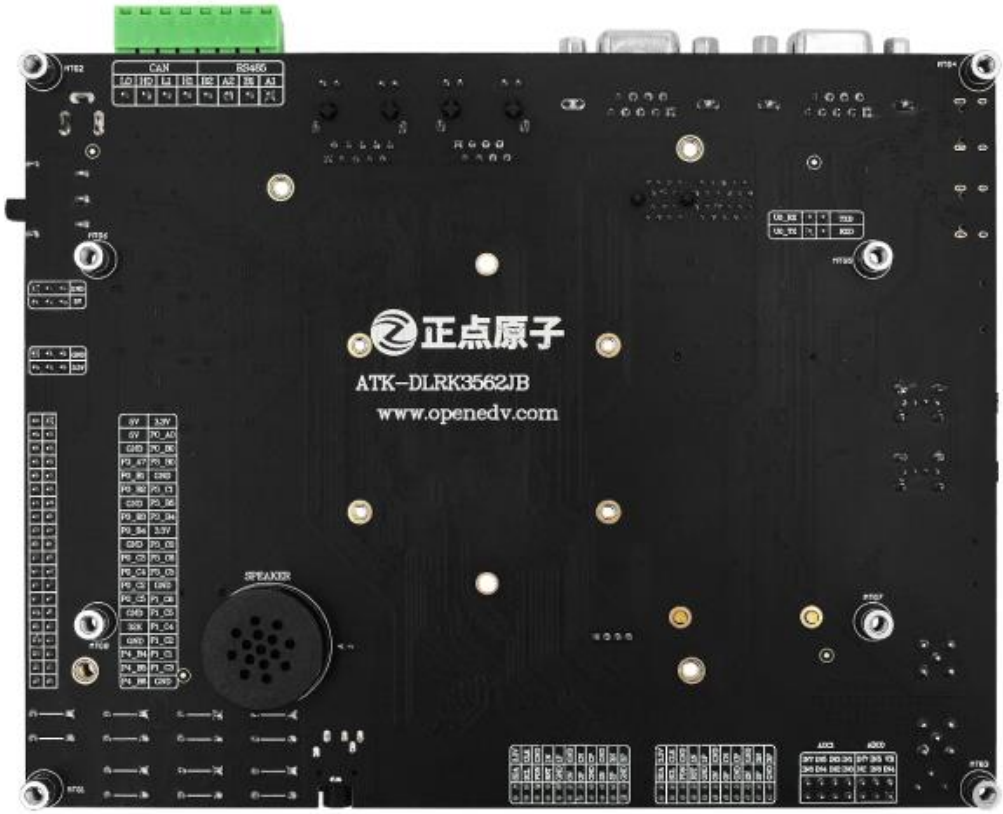


Figure 2.1-3 Back Side of Development Board

Features of ALIENTEK ATK-DLRK3562JB Development Board:

1.Rich Interfaces

The development board provides a variety of standard interfaces, facilitating the experiment and development of various peripherals.

2.Flexible Design

It adopts a core board + baseboard architecture. Most on-board resources can be flexibly configured to meet application requirements under different scenarios.

The core board leads out almost all communication interfaces of the RK3562J processor, including UART, I2C, I2S, SPI, CAN, SDMMC, LVDS / MIPI DSI, MIPI CSI, USB 2.0, USB 3.0 / PCIE 2.1, ADC, 100Mbps Ethernet, Gigabit Ethernet, etc., which greatly facilitates user expansion and application.

3.Rich Onboard Resources

Onboard 100Mbps network card, Gigabit network card, PCIE 2.0 interface, EEPROM memory chip and various interface chips are provided to meet a variety of application requirements.

4.Humanized Design

Each interface is marked with silk screen printing for clear identification; commonly used peripherals are labeled with large silk screen fonts for easy location; interfaces are arranged in a reasonable layout for convenient use.

2.2 Resource Parameters

The parameter block diagram of the RK3562J processor is shown in the figure below.

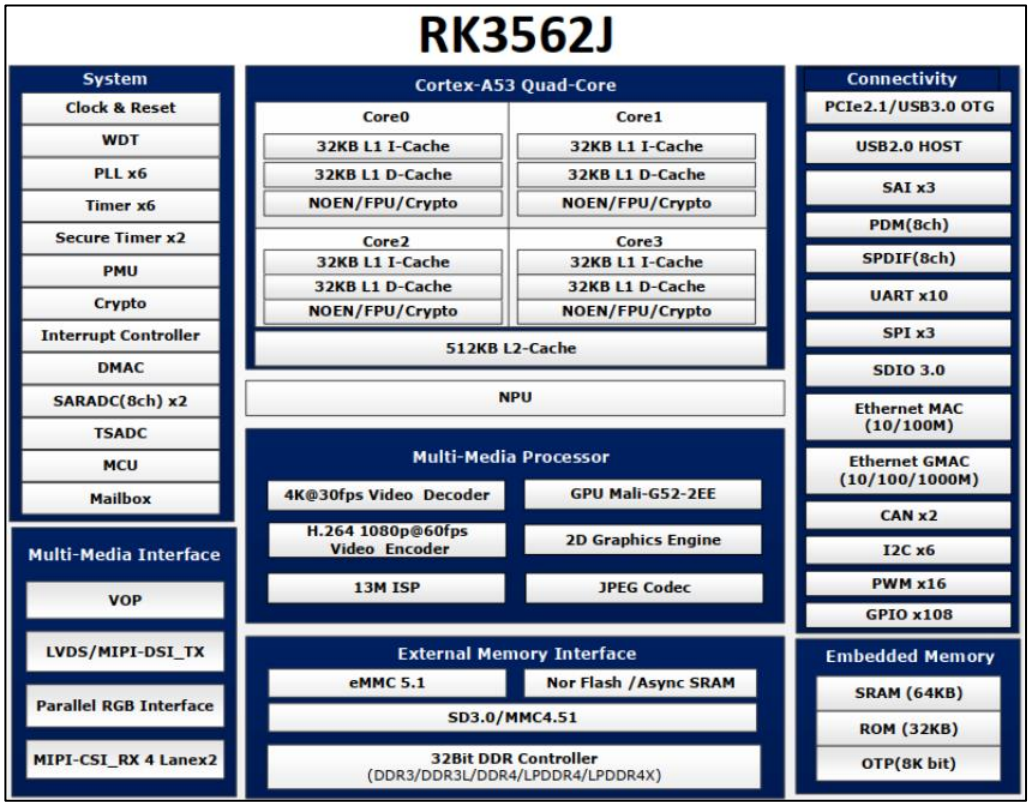


Figure 2.2-1 RK3562J Processor Block Diagram

The hardware resource parameters of the development board are shown in the table below.

Hardware Resources of ATK-DLRK3562JB Development Board		
Core board	CPU	Quad-core 64-bit Cortex-A53, main frequency up to 1.8GHz Main frequency: 1.2GHz in normal mode, 1.8GHz in overdrive mode Single-core Cortex-M0, main frequency 200MHz
	NPU	1.0 TOPS Supports INT8, INT16 and FP16 data types Supports deep learning frameworks such as Caffe, TensorFlow, TF-Lite, ONNX, DarkNet and PyTorch
	GPU	Mail-G52-2EE Supports OpenGL ES 1.1/2.0/3.2, OpenCL 2.0, Vulkan 1.1 and 2D Graphics Engine.
	ISP	13M@30fps, supporting 3A, HDR, 3DLUT and other functions.
	RAM	LPDDR4/4X: 1GB/2GB optional, customizable for mass production.
	ROM	eMMC: 8GB/32GB optional, customizable for mass production.
	PMIC	Dedicated high-efficiency integrated power management chip
	BTB connector	One pair of BTB connector male header, 0.5mm pitch, 2×120P with a total of 240 pins.
Baseboard LEDs	1× Power Indicator	Blue, Power supply status indicator
	1× User Indicator	Red, User programmable LED
	1× Status Indicator	Yellow, Working status indicator for 4G/5G RedCap module
Function Keys	1× RESET Key	System reset key
	1× PWRON Key	Power on/off key
	1× MASKROM Key	System burning key, used to enter MASKROM download mode
	4× User Key	User programmable keys: **V+/RECOVERY**, **V-**, **MENU** and **ESC**. Press the RECOVERY key to enter the LOADER burning mode.
Screen Display	1× MIPI DSI interface	4-lane, supports I2C touch display screen, 2048×1080@60Hz
	1× LVDS interface	Multiplexed with MIPI DSI, only one can be used at a time. 4-lane, supports I2C touch display, 800×1280@60Hz. Supports RGB888/RGB666 and VESA/JEIDA data formats.

Video Input	2× MIPI CSI interface	2-channel 4-lane MIPI CSI interface; it can be split into 4-channel 2-lane MIPI CSI interfaces during custom baseboard design. The total ISP throughput is 13M@30fps.
Wired Communication	2× RS232 Interface	2-channel RS232 interface, DB9 female connector
	2× RS485 Interface	2-channel RS485 interface with chip-level automatic transmit-receive control
	2× CAN Interface	Standard CAN, led out together with RS485 via the same 8PIN terminal block
	1× 100Mbps Ethernet Interface	10/100Mbps adaptive, 100Mbps RJ45 port (with built-in transformer)
	1× Gigabit Ethernet Interface	10/100/1000Mbps adaptive, Gigabit RJ45 port (with built-in transformer)
	2× USB HOST Interface	2-channel USB HOST, USB 2.0, supports connection of USB devices
	1× USB OTG Interface	Type-C interface, supports system flashing, ADB debugging, etc.
	1× UART Interface	Type-C interface, USB to UART, for system debugging
Wireless Communication	1× WIFI/BT Module	RTL8733BU Wi-Fi and Bluetooth combo module, supports Wi-Fi 4 2.4G/5.8G dual-band communication, Bluetooth 5.2, equipped with USB 2.0 interface.
	1× Wi-Fi Antenna Interface	SMA interface for Wi-Fi antenna.
	1× BT Antenna Interface	SMA interface for Bluetooth antenna.
	1× 4G Module Interface	Mini PCIe slot, supports 4G and 5G RedCap modules, with fixing screw holes, communicating via USB 2.0 interface.
	1× SIM Card Interface	Shared by 4G/5G modules, standard Nano SIM card holder.
External Storage	1× PCIE2.0 Interface	PCIE 2.0 36PIN interface, supporting expansion of NVMe M.2 SSD, Gigabit Ethernet, Wi-Fi and other devices.
	1× TF CARD Interface	External storage expansion, standard TF card holder.
	1× SPI FLASH Chip	Adopts SPI communication, 16MB NOR FLASH chip.
	1× EEPROM Chip	Adopts I2C communication, 8KB storage capacity.
Audio Function	1× Headphone Interface	Supports 3.5mm three-section headphone audio playback, output via RK809 chip.
	1× Electret Microphone	On-board microphone, supports audio recording, input to the RK809 chip.

	1× Speaker	Located on the back of the baseboard, 8Ω 1W, driven by the RK809 chip.
Real-Time Clock	1× RTC Real-Time Clock	On-board RTC real-time clock chip on the baseboard, providing external RTC clock.
	1× RTC Button Battery Holder	Compatible with 3V button battery (CR1220), supplies power to the RTC clock chip.
Analog Input	1× Adjustable Potentiometer	10K resistance, used for ADC acquisition with adjustable voltage range of 0~1.8V.
	1× ADC Pin Header Group	2×7PIN ADC pin header group, leading out 12 ADC channels for analog input.
Reserved IO	1× IO Pin Header Group	2×20PIN IO pin header group, leads out multiple IOs with **2.54mm pitch**.
Power Supply	1× Power Toggle Switch	Power on/off toggle switch
	1× 12V Power Input	For powering the development board, 12V DC power input
	1× 5V Power Output	5V power output, 2×3PIN pin header with 2.54mm pitch
	1× 3.3V Power Output	3.3V power output, 2×3PIN pin header with 2.54mm pitch
Others	1× FAN Interface	PWM controlled fan interface

2.3 Software Parameters

The development board supports running different operating systems and a variety of drivers, as shown in the table below.

System and Driver Support		
Operating System	Linux Buildroot (Default factory OS, recommended)	
	Android 14	
	Debian (SDK built-in version, not maintained)	
	RT-Thread, available for running on Cortex-M0 core	
Graphical Interface	QT 5.15	
Driver Support	PMIC	EMMC
	LPDDR4/4X	ISP
	NPU	GPU
	LED	RTC
	KEY	I2C
	CAN	SPI
	I2S	RS485 UART
	RS232 UART	USB HOST
	USB OTG	PCIE 2.0

	USB 3.0	PCIE TO NVMe / Ethernet
	AUDIO OUT	MIC IN
	PWM	W ATCHDOG
	4G	5G RedCap
	WIFI	BLUETOOTH
	MIPI-CSI Camera	Touch Screen
	MIPI-DSI screen	LVDS Display Screen
	100Mbps Ethernet	Gigabit Ethernet
Remark	Provide the complete system driver source code SDK package of the development board, which is easy to use and supports secondary development.	

Chapter 3. Electrical Characteristics

The electrical characteristic parameters of the development board are shown in the table below.

Operating Parameters	Min	Typical	Max
Development Board Operating Voltage	/	12.0V	13.0V
Core Board Operating Voltage 1	/	3.3V	/
Core Board Operating Voltage 2	/	5.0V	5.5V
Development Board Operating Temperature (Commercial Grade)	0°C	/	+70°C
Core Board Operating Temperature (Industrial Grade)	-40°C	/	+85°C

Note: Some chips on the baseboard of the development board are commercial-grade with an operating temperature range of **0 ~ +70°C**. It is recommended to operate within this temperature range to reduce impact on long-term service life.

The power consumption reference values of the development board are shown in the table below.

No.	Test Case	Typical Voltage	Typical Current	Typical Power Consumption
1	Development board powered on, system not running	12V	63mA	0.76W
2	The development board runs only the factory default system without any other external modules connected.	12V	120mA	1.44W
3	The development board is connected with a **5.5-inch 720×1080 MIPI display of ALIENTEK**, running the factory QT interface.	12V	235mA	2.82W

Note: The development board is equipped with abundant interfaces. Its actual power consumption is related to the number of peripheral devices and external modules in use, which shall be measured according to actual application scenarios. The data here is for reference only. For more power consumption test values, please refer to the development board power consumption test document.

Chapter 4. Mechanical Structure

4.1 PCB Board Parameters

Parameter	Core Board	Development Baseboard
PCB Dimensions	50mm*35mm	180mm*140mm
PCB Thickness	1.6mm	1.6mm
PCB Layers	8 layers	4 layers
PCB Process	Immersion Gold	Immersion Gold
Mounting Fixing Holes	4 pcs	8 pcs

4.2 Mechanical Dimensions of Core Board

The ALIENTEK ATK-CLRK3562JB core board is **50mm×35mm**. The mechanical dimension diagram is shown in the figure below.

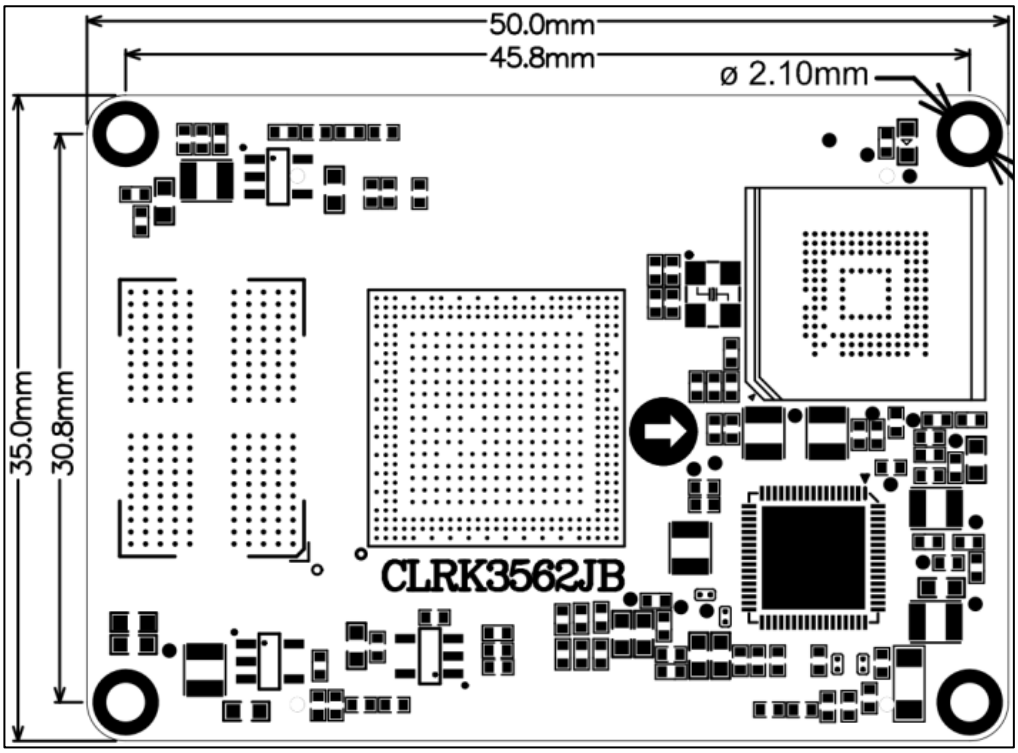


Figure 4.2-1 Mechanical Dimension Drawing of Core Board

Note: The original file path of the above core board mechanical dimension drawing is:

Development Board Network Disk Drive A - Basic Materials\03. Hardware Materials\01.Core Board Materials\04. Core Board Mechanical Dimensions\ATK-CLRK3562JB Core Board Mechanical Dimension Drawing V1.0

4.3 Development Board Mechanical Dimensions

The dimension of the ALIENTEK ATK-DLRK3562JB development board is **180mm×140mm**. The mechanical dimension drawing is shown in the figure below.

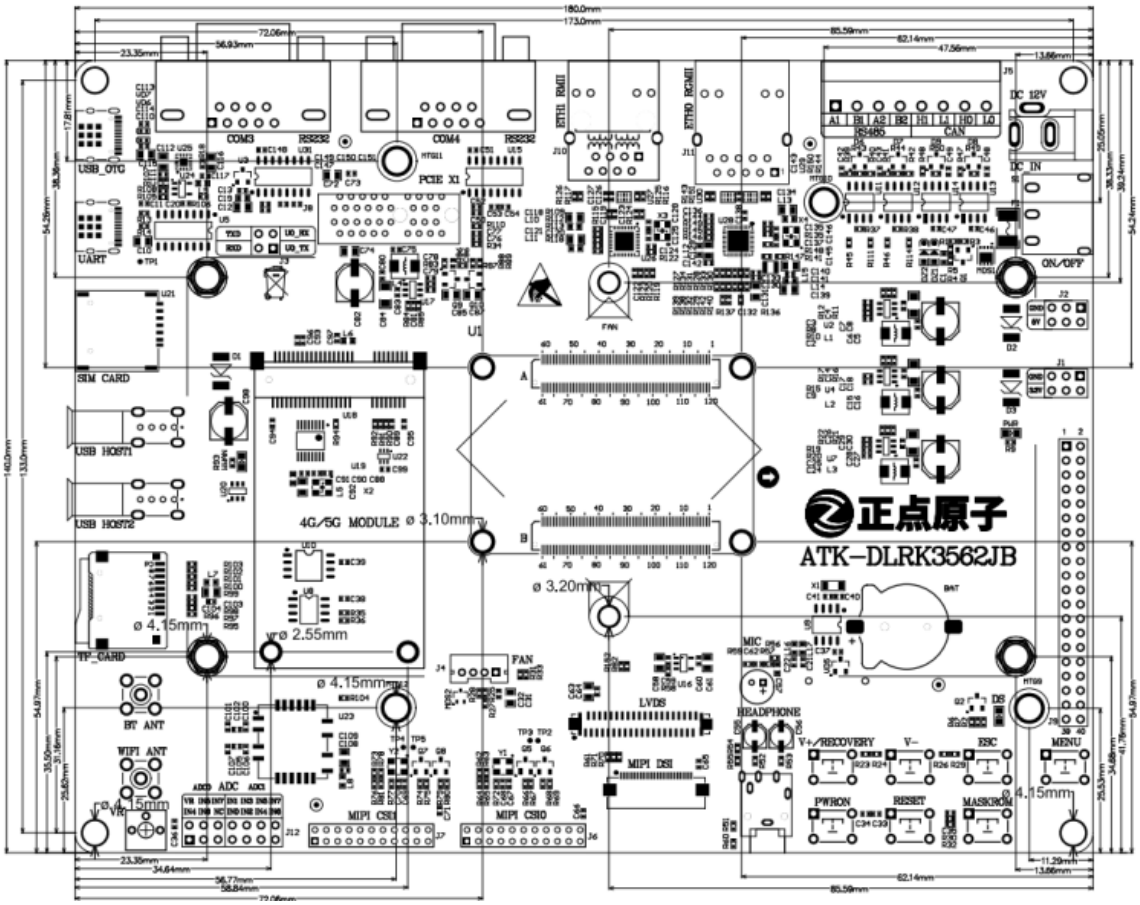


Figure 4.3-1 Development Board Mechanical Dimension Drawing

Note: The original file path of the above development board mechanical dimension drawing is:

Development Board Network Disk Drive A - Basic Documents\03. Hardware Documents\02.Development Board Documents\04. Baseboard Mechanical Dimensions\ATK-DLRK3562JB Development Board Mechanical Dimension Drawing V1.0

Chapter 5. Development Board Resources

ALIENTEK provides abundant development resources for the ATK-DLRK3562JB development board.

Software resources include but are not limited to U-Boot source code, Linux kernel source code, peripheral driver source code, file system source code, Qt interface source code, and Linux application source code, as well as related development tools and development environments.

Documentation resources consist of detailed development documents and auxiliary materials, including dozens of manuals such as the SDK Development Manual, Embedded Linux Driver Development Guide, Linux Application & Hardware Development Examples, and Embedded AI Development Manual. The contents are easy to understand and highly practical.

Hardware resources cover the core board specification, hardware reference manual, minimal system hardware design reference, baseboard schematic and original project files, core board interface datasheet, baseboard component footprint library, mechanical dimension drawings, and chip reference manuals. The comprehensive hardware materials fully meet the development requirements of engineers.

All supporting resources of the development board are available for download via cloud disk.

5.1 Document Download

ATK-DLRK3562JB Development Board Resource Download Center:
http://www.openedv.com/docs/boards/arm-linux/RK3562J%20Linux.html

5.2 Resource Description

First-Level Directory of Development Board Documents:

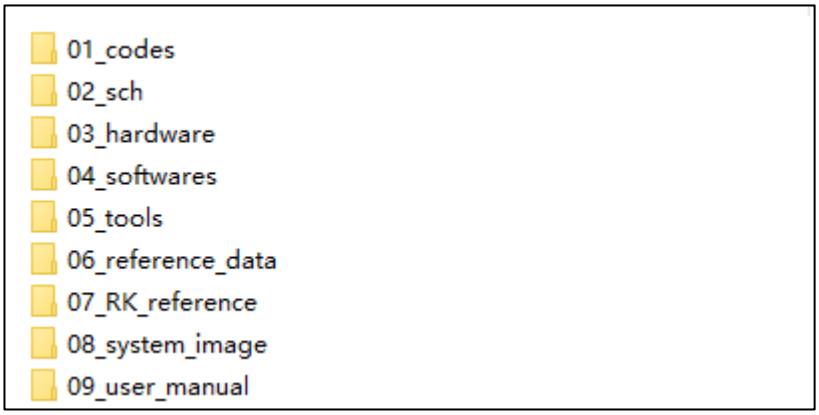


Figure 5.2-1 First-Level Directory of Development Board Documents (Disk A)

Figure 5.2-2 First-Level Directory of Development Board Documents (Disk B)

Directory Description of Development Board Resources (Disk A):

Disk A Directory	Description
01_codes	AI routines, heterogeneous core routines, Linux driver routines, QT interface programs, application and hardware development examples, etc.
02_sch	Development board baseboard schematic

03_hardware	Hardware documents of development baseboard and core board, including specifications, on-board chips, baseboard footprint library, mechanical dimensions, connector specifications, baseboard schematic project files, test reports and other detailed and practical materials.
04_softwares	Tools for development environment setup, Ubuntu virtual machine, serial port debugging tools, USB drivers, etc.
05_tools	ADB debugging tools, Rockchip official tools, cross-compilation tools, etc.
06_reference_data	Chip datasheets, protocol manuals, tutorial reference manuals, etc.
07_RK_reference	Official documents of Rockchip SDK
08_system_image	Factory default system image package
09_user_manual	Complete and detailed supporting development documents to help users develop quickly

Directory Description of Development Board Resources (Disk B):

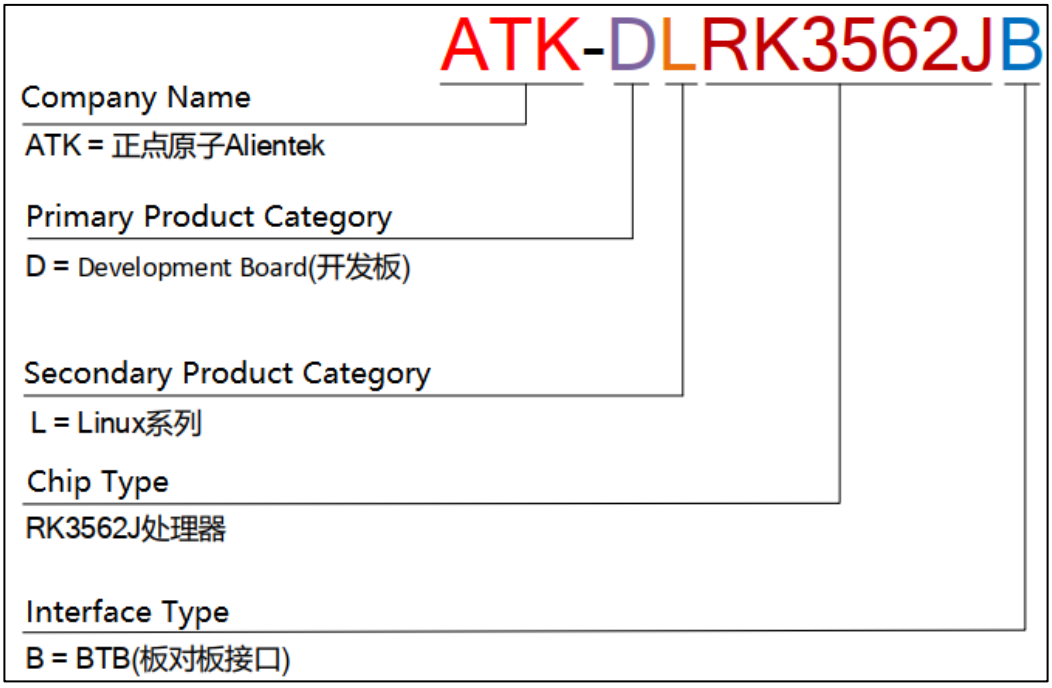
Disk B Directory	Description
01_ Ubuntu_development	Ubuntu virtual machine, applicable to development board SDK source code development and other scenarios.
02_SDK	Factory System & SDK Source Code Package of Development Board

Note: Due to limited space, the contents of each directory are not elaborated here. Please download the development board resources for detailed reference.

Chapter 6. Development Board Model

Development Board Model: ATK-DLRK3562JB

The model parameter analysis is as follows.



Chapter 7. Development Board Compatible Modules

The factory default system of the development board is compatible with and supports ALIENTEK camera modules, MIPI displays, LVDS displays, 4G module EC20, and other peripherals.

Other modules are not sold by ALIENTEK; additional modules can be debugged and adapted according to actual application scenarios.

Chapter 8. Precautions and maintenance

Notes

- Do not plug and unplug peripheral modules with power!
- Before using the product, please carefully read this manual and related development manuals, and pay attention to the applicable matters of the platform.
- Follow all instructions and warnings on the product.
- Please use this product in a cool, dry and clean place.
- Please keep the product dry. If any liquid splashes or soaks, power off immediately and let dry thoroughly.
- Do not use organic solvents or corrosive liquids to clean the product.
- Do not use or store this product in dusty, dirty and messy environment.
- If not used for a long time, please package this product, pay attention to moisture-proof and dust-proof.
- Pay attention to the ventilation and heat dissipation of the product during use to avoid component damage caused by excessive temperature during operation.
- Do not use this product in alternating hot and cold environment to avoid dew damage to components.
- Do not treat this product roughly, drop, knock or shake violently may damage the line and components.
- Pay attention to anti-static when using this product.
- FPC flexible cable is fragile, when plugging cable, pay attention to check whether the metal at both ends of the cable is misplaced and falling off.
- All products have passed the product test before shipment. Please use the development board corresponding to the ALIENTEK for power on test for the first time.
- Do not repair or disassemble the company's products by yourself. If the product fails, please contact the company in time for maintenance.
- Unauthorized modification or use of unauthorized parts may damage the product, the resulting damage will not be repaired.

Chapter 9. After sales service

9.1 Terms of after-sales service

1). After receiving the goods, please open them in front of the express, and sign after acceptance. If you find that the goods are less after signing, take photos in time and contact the seller's customer service to explain the situation within 15 days. If the feedback is lack of goods after 15 days, we will not reissue the goods. Other reasons notwithstanding).

2). 15 days -1 month: we are responsible for the return freight repair of product problems. Human factors damage expensive main chip or LCD screen, touch screen. The buyer needs to pay the cost and one time shipping fee, no maintenance fee.

3). 1-3 months: the problem of the product itself (non-human factors), we are responsible for the delivery of the past freight maintenance. If the main chip is burned out and the LCD screen and touch screen are damaged, the buyer needs to pay the cost, and the maintenance fee is not charged.

4) After 3 months: the buyer shall bear the return freight and the cost of chip, LCD screen and touch screen. No service charge.

9.2 After-sales Support

Technical support: Contact the customer service to obtain the group number.

QQ group: ALIENTEK Rockchip Communication group: 1079854944
ALIENTEK RK3562J Technical Support Group:
1072910286 (order number required)

Taobao shop: ALIENTEK flagship store

Forum: http://www.openedv.com/forum.php